

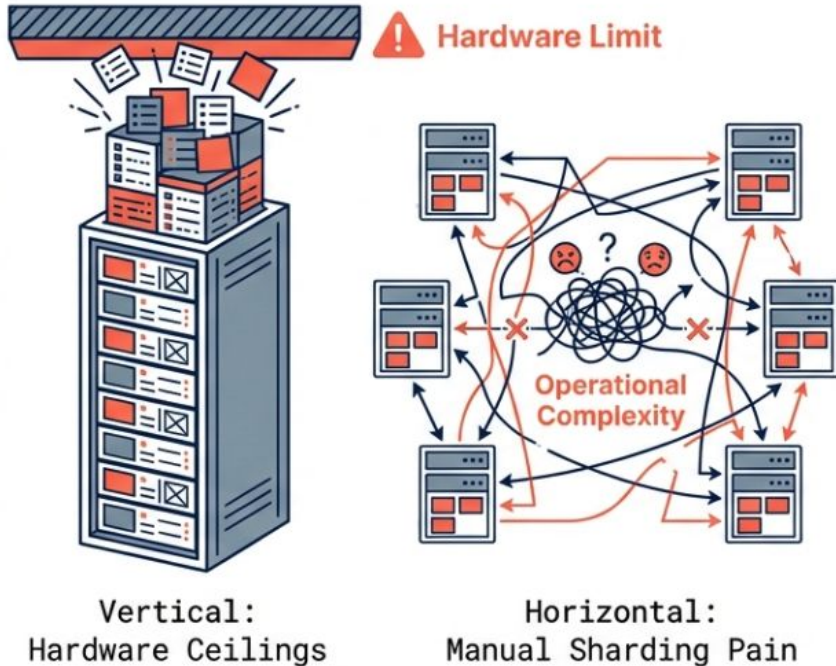


TiDB

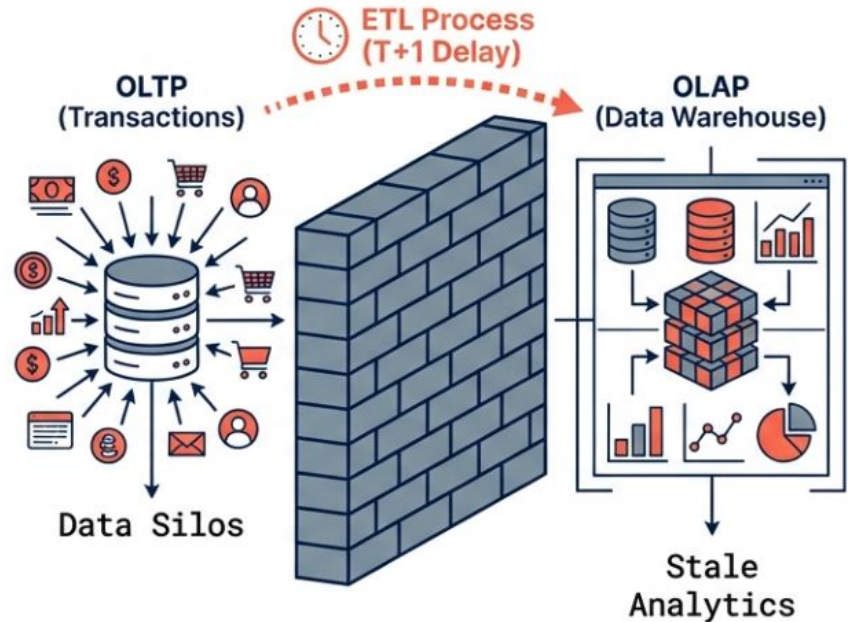
Modern Database Architecture for Real-Time Workloads

Why new database

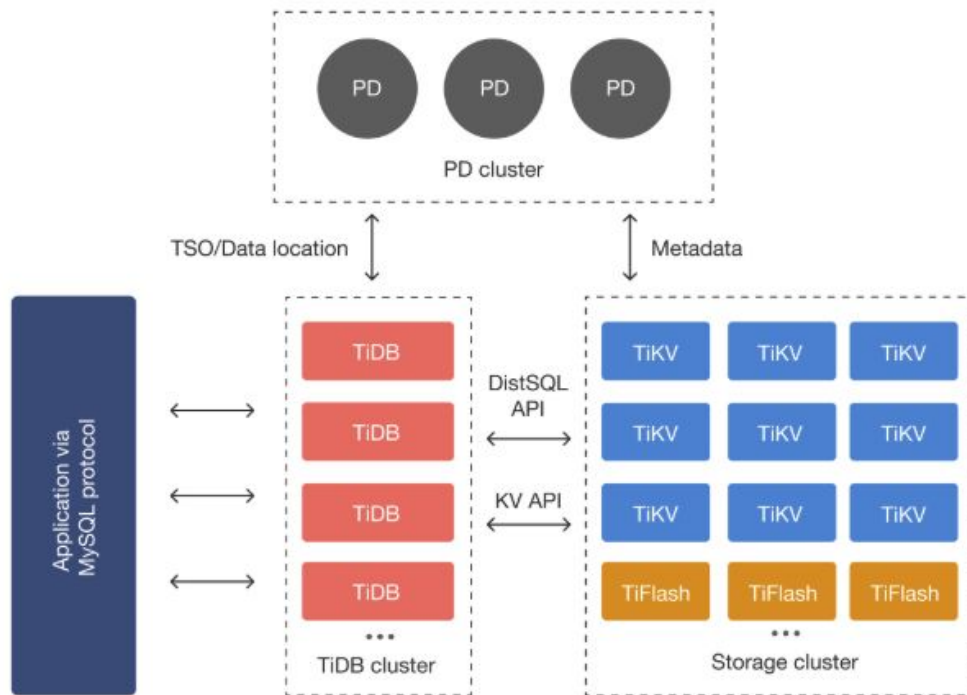
The Scaling Problem



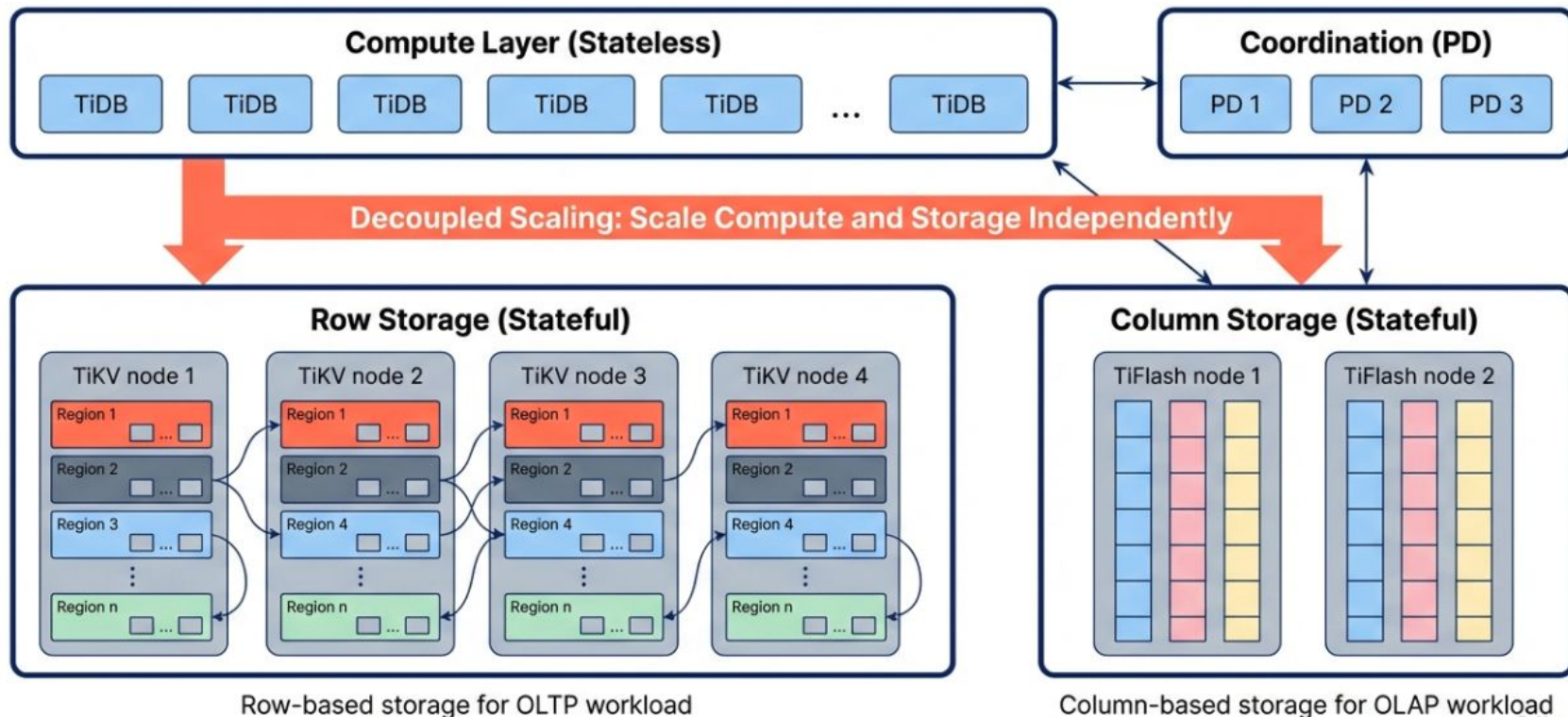
The Workload Wall



Architecture

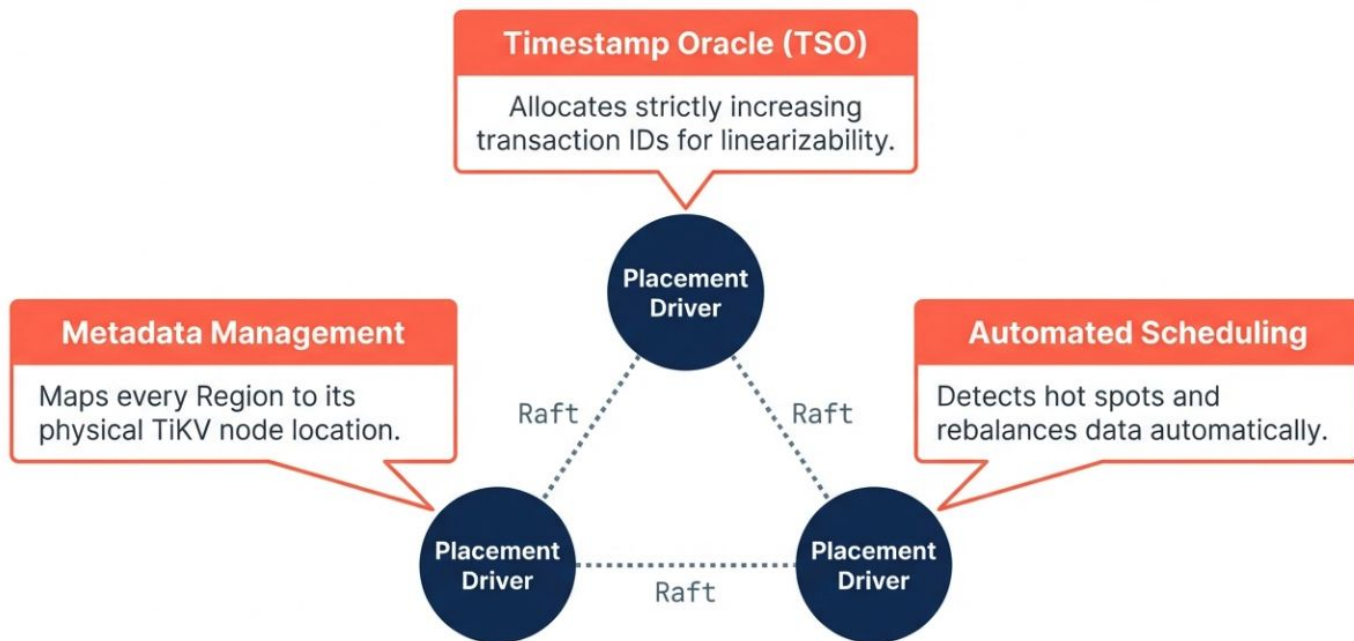


Architecture



Placement Driver (PD)

The Coordinator (PD): The Cluster's Nervous System



TiDB Server

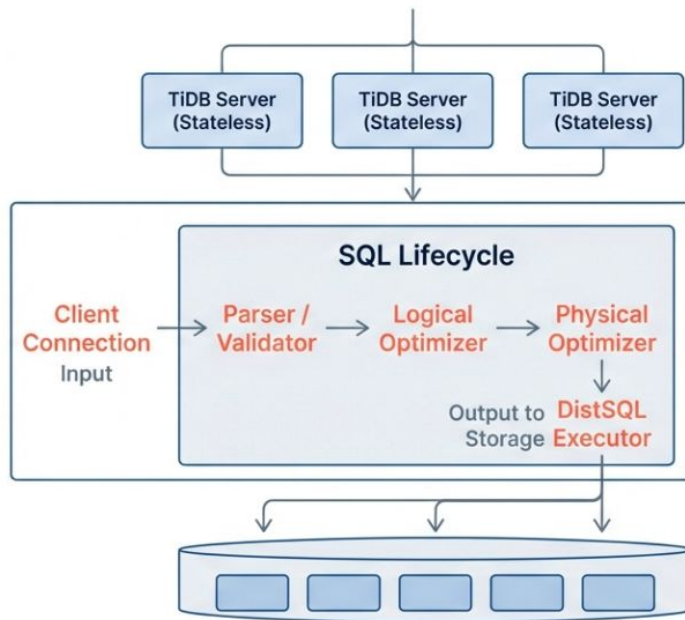
The SQL Layer (TiDB Server): The Stateless Brain

Protocol: MySQL 5.7 Compatible

Parsing & Compiling: AST
Generation

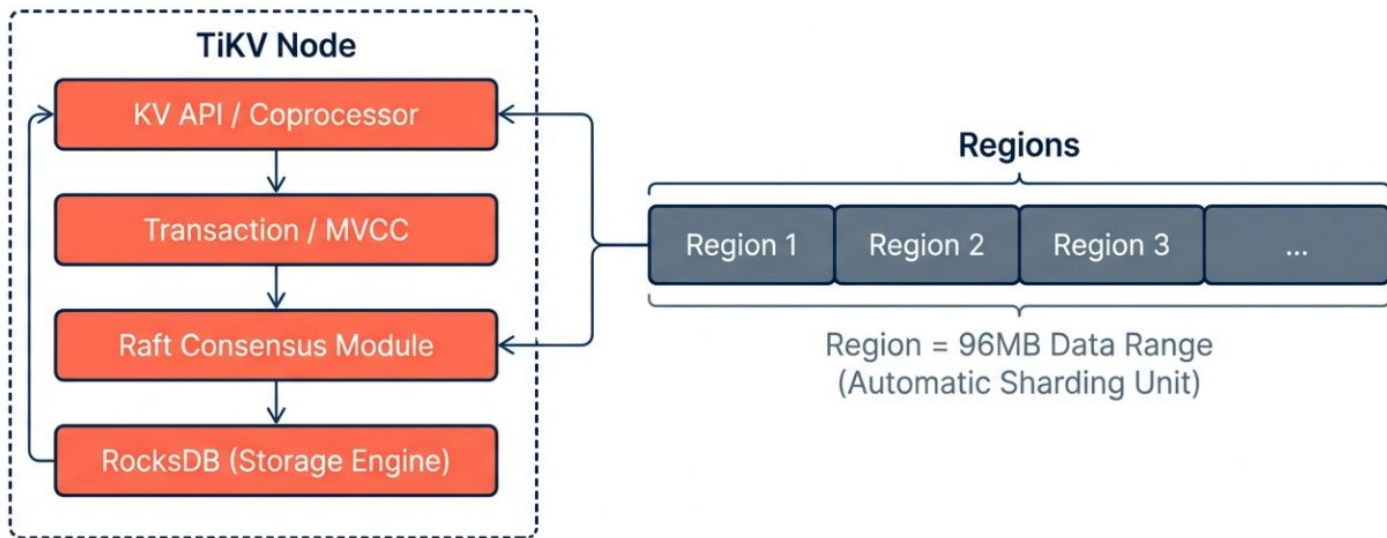
Translation: Relational SQL →
Key-Value Requests

Scaling: Stateless (Add nodes
instantly)



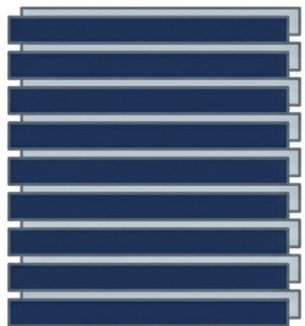
TiKV

The Storage Layer (TiKV): The Distributed Heart

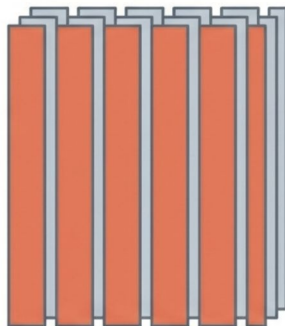


TiFlash

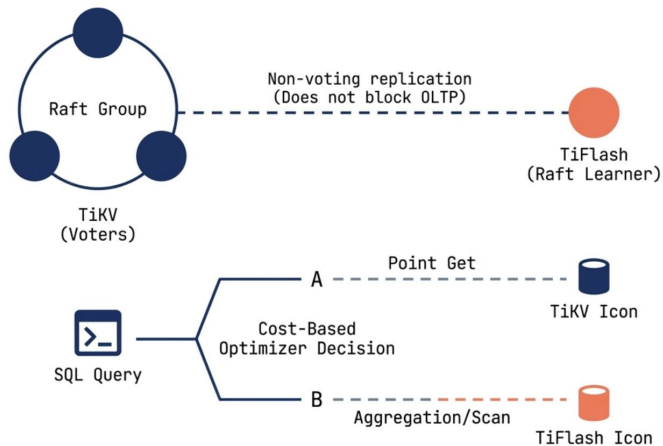
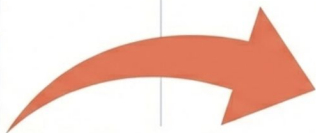
TiKV (Row Store)



TiFlash (Column Store)



Real-time Asynchronous Replication.





Trade-offs



Pros

- Horizontal Scaling
- HTPA
- Good documentation
- Build modern language, Go, Rust
- K8S operator
- TiCDC
- TiDB Cloud

Cons

- Complex infra setup
- Hostspots
- Advanced configuration for throughput (SHARD_ROW_ID_BITS)
- No triggers and stored procedures

Comparisons



Feature	TiDB	MySQL + Vitess	YugabyteDB	CockroachDB
SQL Compatibility	MySQL	MySQL	PostgreSQL	PostgreSQL
Horizontal Scaling	Native automatic scaling	Sharding-based scaling	Native automatic scaling	Native automatic scaling
Distributed ACID Transactions	Yes	Limited / complex	Yes	Yes
Sharding	Automatic	Manual / configured	Automatic	Automatic
Multi-Region Support	Yes	Difficult to implement	Yes	Very strong
Consistency Model	Strong consistency	Depends on configuration	Strong consistency	Strong consistency
Architecture	SQL layer + distributed KV storage	Proxy/router + MySQL shards	PostgreSQL layer + distributed KV	Integrated SQL + distributed KV
Storage Engine	TiKV	InnoDB (MySQL)	DocDB	RocksDB
OLTP Workloads	Excellent	Excellent	Excellent	Excellent
OLAP / Analytics	Good (TiFlash column store)	Weak	Moderate	Moderate
Operational Complexity	Medium	High	Medium	Medium
Ecosystem	Growing	Very large (MySQL ecosystem)	Medium	Large



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